Group Member: I am working on it alone
Discussion Logbook: N/A as I am working on it alone
Distribution of work: N/A
Github Link: https://github.com/jtclayto23/CSE-490-Project
Design plan:
https://docs.google.com/document/d/18KkAAUpPk_yrXcSqe1a7dfYI04fljVzXXlE4Jq6lSzQ/edit?usp=sharing

I have completed the diagram of the entire processor (included in GitHub), and designed each component. I worked on the register file, PC, ALU, Control unit, Data Memory component, and developed the logic for branching and jumping. I developed the individual MUX, sign extension, SLL, AND, and ADD components for the pipeline.

I also finished designing all of the control signals that will be used in the project. They will include the common ones like mem_write, mem_read, reg_write, etc, but will also include a special control signal for JMP instructions and branch instructions. It will also include a BNE signal that wires into a XOR gate, along with a Zero flag from the ALU, to make sure the program interprets the Zero flag correctly depending on if the branch instruction is a BEQ or BNE.

Lastly, I finished coding the logic for all the instructions in the program, not just the R-type anymore. The next couple of weeks I will finish up wiring them all together and begin heavily testing it, including on a physical FPGA board instead of just Verilog simulations. I will also extensively document how every component works, and make a more professional-looking design plan once everything is finalized. All files are in my GitHub repository.